

Project 06: AI Policy & Compliance Governance Engine

Hybrid Rule & LLM Corporate Expense Governance

1. Executive Overview

The AI Policy & Compliance Engine evaluates corporate expense claims against financial policy rules. It uses a deterministic rule engine to flag maximum transaction limits (\$1,000), restricted merchants, and missing receipts (\$50 threshold), combined with Gemini 2.5 Flash to generate formal executive compliance reports.

2. System Architecture & Workflow Pipeline

- Deterministic Rule Engine Checks (Transaction Limits, Merchant Restrictions, Receipts)
- Status Classification: COMPLIANT / WARNING / NON_COMPLIANT
- Google Gemini 2.5 Flash Auditor -> Formal Executive Audit Summary
- Human Oversight Flagging for NON_COMPLIANT and WARNING claims

3. API Endpoints Specification

Method	Endpoint	Description
GET	/	Serves Policy Compliance web UI
GET	/api/policies	Returns corporate policy rules and threshold definitions
POST	/api/audit-claim	Audits expense claim and returns status, violations, and summary

4. Deployment & Automated Verification

Standalone Uvicorn Launch:

```
cd 06_policy_compliance/src && python -m uvicorn server:app --host 127.0.0.1 --port 8006
```

Automated Test Suite Execution:

```
python -m pytest 06_policy_compliance/tests/
```